

Problem Set 10

Due Tuesday, Dec 31, 2024

Optimal Linear Decoder

Consider a linear decoding framework, where

$$\begin{aligned} s_+ &= \sum_{i=1}^N w_i r_+^i \\ s_- &= \sum_{i=1}^N w_i r_-^i \end{aligned} \tag{1}$$

Given the noise covariance matrix Σ , where $\Sigma_{ij} = \langle \delta r_i \delta r_j \rangle$, where the average is over the same stimulus trials. Show that the optimal linear decoder that maximizes the Signal-to-Noise ratio

$$d^2 = \frac{(\langle s_+ \rangle - \langle s_- \rangle)^2}{\frac{1}{2}(\delta s_+^2 + \delta s_-^2)} \tag{2}$$

is given by

$$\mathbf{w} \simeq \Sigma^{-1} \Delta \boldsymbol{\mu}, \tag{3}$$

And the corresponding SNR is

$$d^2 = \Delta \boldsymbol{\mu}^T \Sigma^{-1} \Delta \boldsymbol{\mu} \tag{4}$$

where $\Delta \boldsymbol{\mu} = \vec{\mu}_+ - \vec{\mu}_-$, and $\vec{\mu}$ is the mean population response vector to a given stimulus.

Linear Fisher Information

Instead of considering two discrete stimuli (+ or -), now consider the population mean response vector with respect to a continuous stimulus s

$$\mathbf{m}(s) = [m_1(s), m_2(s), \dots, m_N(s)]^T.$$

Let us assume that the likelihood distribution follows the multidimensional Gaussian distribution,

$$P[\mathbf{r}|s] = \frac{1}{\sqrt{(2\pi)^N \det \Sigma}} \exp \left(-\frac{1}{2} [\mathbf{r} - \mathbf{m}(s)]^T \Sigma^{-1} [\mathbf{r} - \mathbf{m}(s)] \right) \quad (5)$$

Show that the Fisher information is a generalization of Eq. 4

$$I_F = \mathbf{m}'^T(s) \Sigma^{-1} \mathbf{m}'(s) \quad (6)$$

Fisher Information of independent Poisson Neurons encoding high-dimensional stimulus

In the class, we derived the following formula (also see lecture notes)

$$I_F = T \sum_{i=1}^N \frac{m'_i(s)^2}{m_i(s)}, \quad (7)$$

Now consider a Gaussian tuning curve with

$$m_i(s) = r_{max} \exp \left[-\frac{1}{2} \left(\frac{s - s_i}{2\sigma_r^2} \right)^2 \right] \quad (8)$$

which leads to

$$I_F = T \sum_{i=1}^N \frac{r_{max}(s - s_i)^2}{\sigma_r^4} \exp \left[-\frac{1}{2} \left(\frac{s - s_i}{2\sigma_r^2} \right)^2 \right] \quad (9)$$

This sum can be approximated by an integral over their preferred stimulus values s_i and multiplied by a density ρ_s such that $\rho_s \Delta s$ represents the number of neurons with a preferred stimulus between s_i and $s_i + \Delta s$. Compute this integral and show what you get. Generalize this model to multidimensional stimuli where

$$\mathbf{s} = [s_1, s_2, \dots, s_D]^T,$$

and use precisely the same Gaussian tuning curve for each stimulus dimension, and show what you get and discuss how the results depend on the tuning with σ_r^2 .